

Two-Link Robot Arm: Kinematics and Control Loop Design

Determining end-effector position from joint coordinates, and a two-loop (joint-level + task-level) controller architecture

1. Problem Setup

A two-link planar robot arm has two rigid links of length $L1$ and $L2$, connected by two rotary joints, each driven by a motor. The base joint rotates by angle θ_1 , measured from the fixed x-axis. The elbow joint rotates by angle θ_2 , measured relative to the direction of link 1 (this is the standard convention because it keeps the equations simple and independent of link 1's absolute orientation). The tip of link 2 is called the end effector, and its position is written as the coordinate (x, y) .

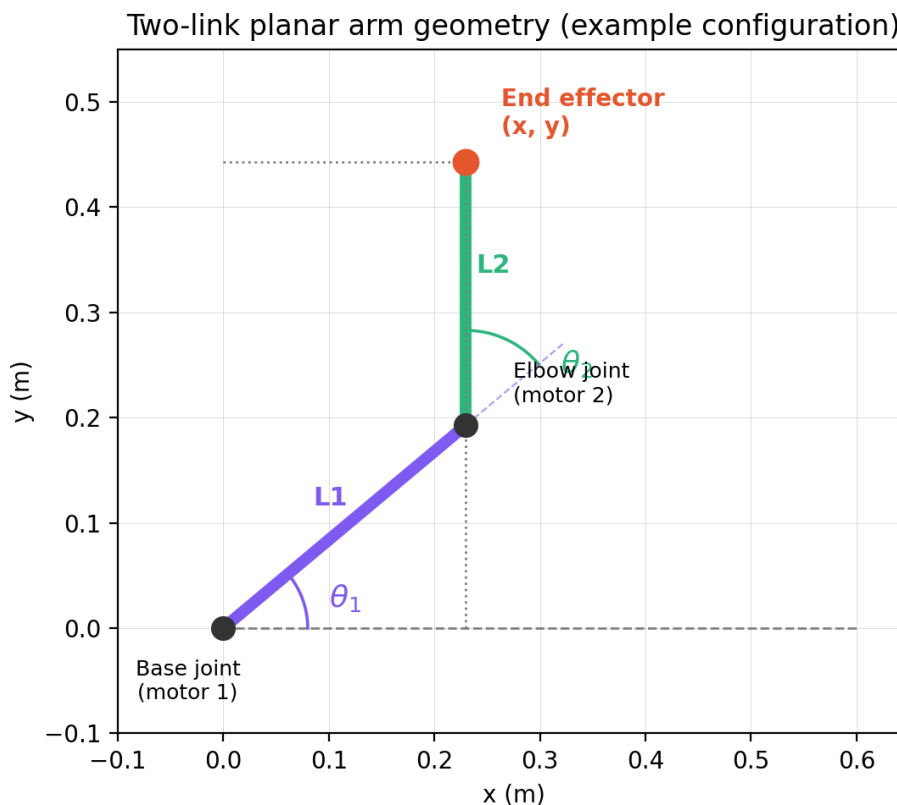


Figure 1: Geometry of the two-link arm, with an example configuration used throughout this document.

2. Forward Kinematics — From Joint Angles to End-Effector Position

Forward kinematics answers the question: given the two joint angles, where is the tip of the arm? This is the natural starting point, because the control loop design later in this document depends on being able to compute the actual end-effector position from the

actual measured joint angles.

Step 1 — Locate the elbow (end of link 1)

Link 1 is a simple rotating rod of length $L1$ at angle $\theta1$ from the x-axis. Resolving it into horizontal and vertical components gives the elbow position:

$$x_{\text{elbow}} = L1 \cdot \cos(\theta1) \quad y_{\text{elbow}} = L1 \cdot \sin(\theta1)$$

Step 2 — Find link 2's absolute angle

$\theta2$ is measured relative to link 1, not relative to the ground. Since link 1 itself is already tilted by $\theta1$, link 2's true angle with respect to the fixed x-axis is the sum of the two:

$$\theta_{\text{link2, absolute}} = \theta1 + \theta2$$

Step 3 — Add link 2's contribution to reach the end effector

Starting from the elbow, link 2 extends further by length $L2$ at its absolute angle ($\theta1 + \theta2$). Adding this to the elbow position gives the full forward kinematics equations:

$$x = L1 \cdot \cos(\theta1) + L2 \cdot \cos(\theta1 + \theta2) \\ y = L1 \cdot \sin(\theta1) + L2 \cdot \sin(\theta1 + \theta2)$$

These two equations are the complete forward-kinematics solution: plug in any pair of joint angles and they return exactly where the tip of the arm is.

2.1 Worked Numerical Example

Assume a simple configuration to make the steps concrete:

Quantity	Value
Link 1 length, $L1$	0.30 m
Link 2 length, $L2$	0.25 m
Base joint angle, $\theta1$	40 deg
Elbow joint angle, $\theta2$	50 deg

Step A — elbow position:

$$x_{\text{elbow}} = 0.30 \cdot \cos(40^\circ) = 0.30 \cdot 0.766 = 0.230 \text{ m} \\ y_{\text{elbow}} = 0.30 \cdot \sin(40^\circ) = 0.30 \cdot 0.643 = 0.193 \text{ m}$$

Step B — absolute angle of link 2:

$$\theta1 + \theta2 = 40^\circ + 50^\circ = 90^\circ$$

Step C — end-effector position:

$$x = 0.230 + 0.25 \cdot \cos(90^\circ) = 0.230 + 0 = 0.230 \text{ m} \\ y = 0.193 + 0.25 \cdot \sin(90^\circ) = 0.193 + 0.25 = 0.443 \text{ m}$$

So for this configuration, the end effector lands at $(x, y) \approx (0.230 \text{ m}, 0.443 \text{ m})$, which is exactly the point marked in Figure 1. This is the core forward-kinematics calculation — the same two equations, evaluated with whatever joint angles the motors have actually reached, will be reused directly in the control loop in Section 3.

Note: inverse kinematics (going the other way — from a target (x, y) to the joint angles needed to reach it) uses the law of cosines on this same geometry, and is what the inverse kinematics block in Figure 2 performs.

3. Control Loop Design

The controller has three loops: one PID loop for joint 1, one PID loop for joint 2, and a final outer loop that checks whether the end effector actually reached the target position. The outer loop exists because each joint's small tracking error, once passed through the sine/cosine relationships in forward kinematics, can combine into a noticeably larger end-effector position error — so the joints reaching their target angles does not by itself guarantee the tip is in the right place.

3.1 Joint 1 Loop (inner loop, base joint)

A PID controller compares the desired angle $\theta_{1_desired}$ (from inverse kinematics) against the actual angle θ_{1_actual} (read from motor 1's encoder), and drives motor 1 to close that gap:

$$\begin{aligned} e1 &= \theta_{1_desired} - \theta_{1_actual} \\ u1 &= Kp1 \cdot e1 + Ki1 \cdot \int e1 \, dt + Kd1 \cdot (de1/dt) \end{aligned}$$

$u1$ is the control signal (motor torque or voltage command) sent to motor 1. This is the same structure used in your earlier 1-link stiffness/damping/force model — here it is applied independently to joint 1.

3.2 Joint 2 Loop (inner loop, elbow joint)

Identically, joint 2 gets its own independent PID loop comparing desired vs. actual θ_2 :

$$\begin{aligned} e2 &= \theta_{2_desired} - \theta_{2_actual} \\ u2 &= Kp2 \cdot e2 + Ki2 \cdot \int e2 \, dt + Kd2 \cdot (de2/dt) \end{aligned}$$

Joints 1 and 2 run these loops in parallel, each with its own gains, since the two joints generally carry different loads and inertias.

3.3 Outer Loop (final position check)

Even with both joint loops converged, take the actual measured angles and pass them through the forward-kinematics equations from Section 2 to find where the tip really is, then compare that to the original target:

$$\begin{aligned} (x_{actual}, y_{actual}) &= \text{ForwardKinematics}(\theta_{1_actual}, \theta_{2_actual}) \\ e &= (x_{target} - x_{actual}, y_{target} - y_{actual}) \end{aligned}$$

If the position error e is larger than an acceptable tolerance, it is converted into a small joint-angle correction using the Jacobian matrix $J(\theta)$, which relates small joint changes to small end-effector changes:

$$\Delta\theta = J^{-1}(\theta) \cdot e$$

This $\Delta\theta$ is added on top of the desired angles going into the two joint PID loops, and the cycle repeats: joints move, actual angles are re-measured, forward kinematics recomputes the tip position, and the error is rechecked. This continues until the error falls within tolerance, at which point the arm is confirmed to be at the correct position — not just that the joints reached some angle.

Control architecture: Joint 1 loop + Joint 2 loop + final outer-loop position check

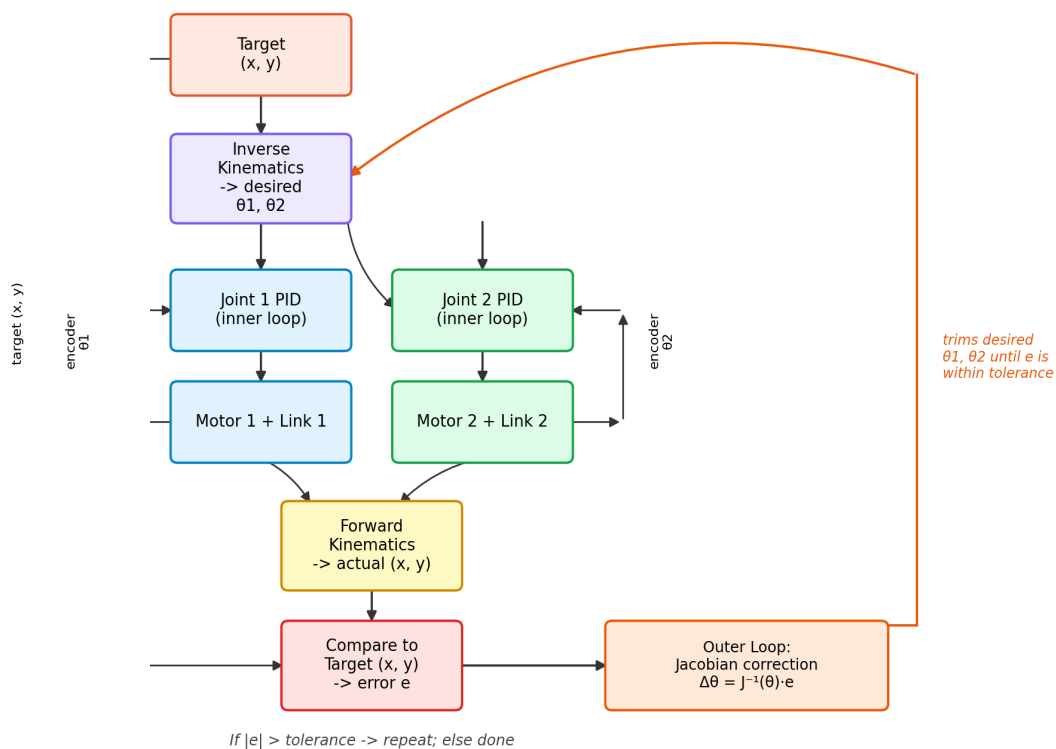


Figure 2: Full control architecture — Joint 1 and Joint 2 inner PID loops, plus the outer Jacobian-based loop that verifies and corrects the final end-effector position.

3.4 Summary of the Three Loops

Loop	Compares	Corrects	Runs on
Joint 1 (inner)	θ1 desired vs actual	Motor 1 torque/voltage	Every control cycle
Joint 2 (inner)	θ2 desired vs actual	Motor 2 torque/voltage	Every control cycle
Outer (final)	(x,y) target vs actual	Desired θ1, θ2 (via Jacobian)	After joints settle

In short: the two inner loops make sure each motor reaches its commanded angle; the outer loop makes sure that, together, those two angles actually put the end effector where it was meant to go — and trims the commands if not.